

Data Structure

Course No. 40989

Week #13

Balanced BST (AVL Tree & Red-Black Tree)

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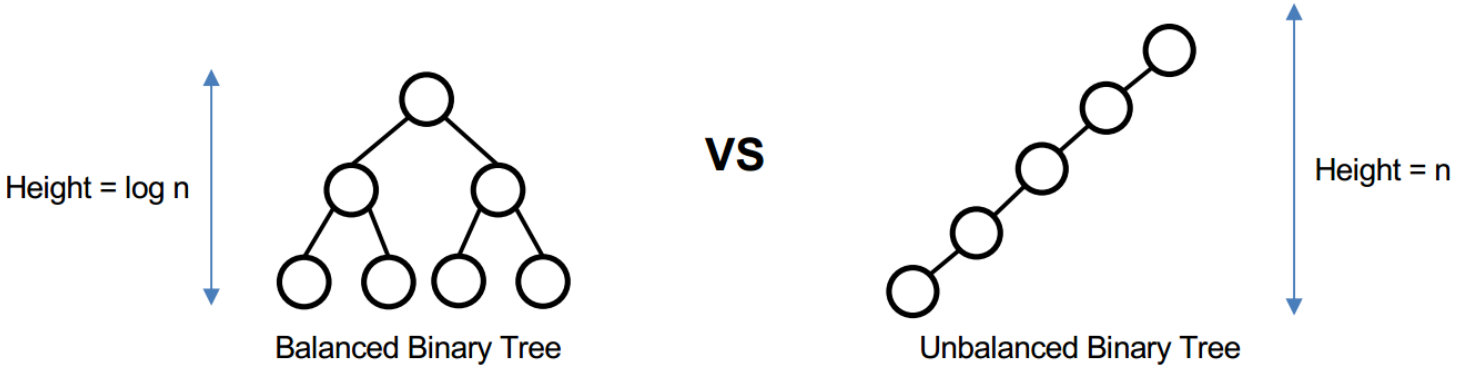
Office : 310-430

AVL Tree

Motivation

Drawbacks of Binary Search Tree (BST)

- The performance of operations on BST depends on its **height**
- **However,** the best case is O(log n) while the worst case is O(n)
- **There exists "bad" sequences that make BST unbalanced.**



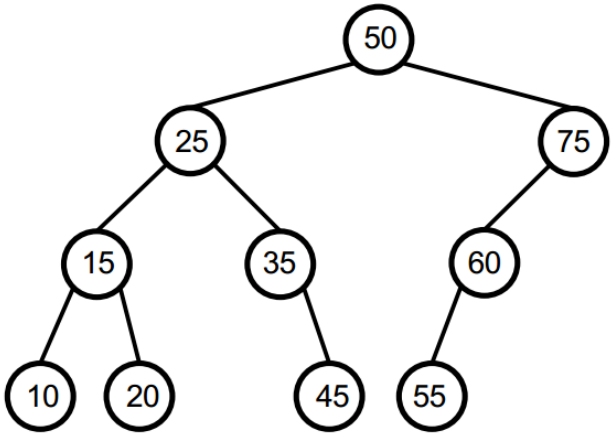
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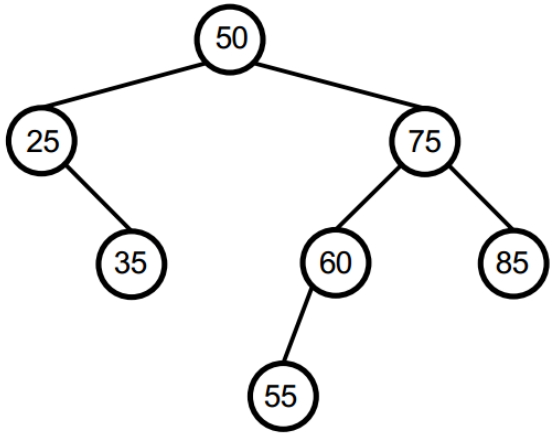
AVL Tree

Definition

Adelson-Velsky and Landis (AVL) Tree

- A self-balancing BST in which the difference between the heights of left and right subtrees of any nodes is at most 1, ensuring logarithmic tree height.





Are they AVL trees?
How can we check if a (sub)tree is balanced or not?

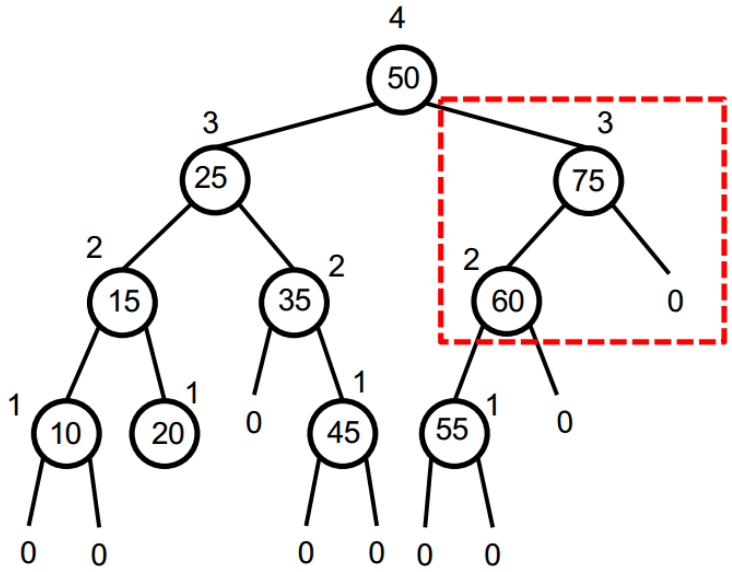
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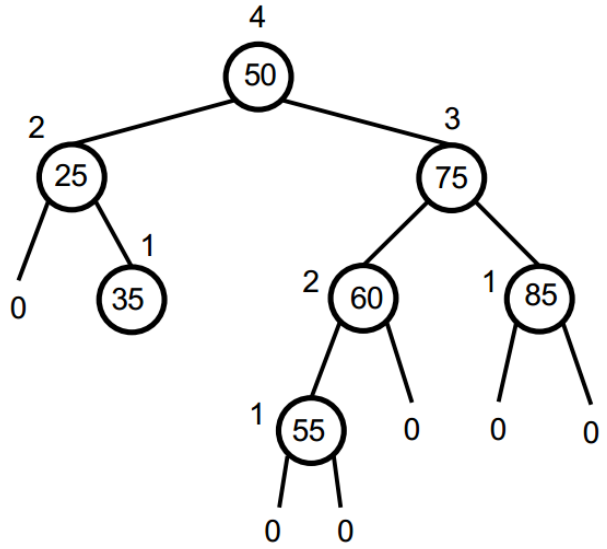
AVL Tree

Definition

Adelson-Velsky and Landis (AVL) Tree

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How can we check if a (sub)tree is balanced or not?

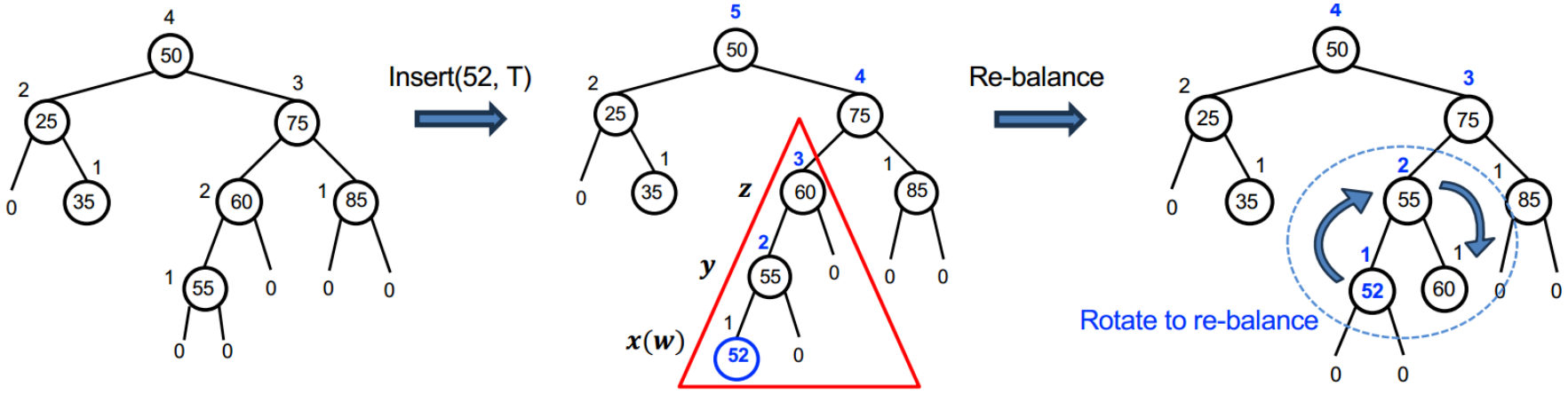
→ **Balance Factor:** the difference between the heights of its left and right subtrees

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Operations: Insertion

Inserting a key w into T

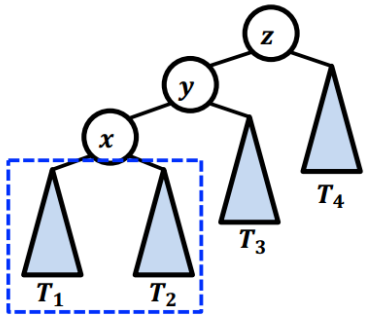
- [1] Perform standard BST insertion for the new key w
- [2] Starting with w, find the first unbalanced node (i.e., one of the ancestors)
  - Let z be the first unbalanced node, y and b be the child and grandchild of z
  - y and x come on the path from w to z (x is possibly w itself)
- [3] Re-balance the tree by performing appropriate rotations on the subtree rooted at the z, the unbalanced node.



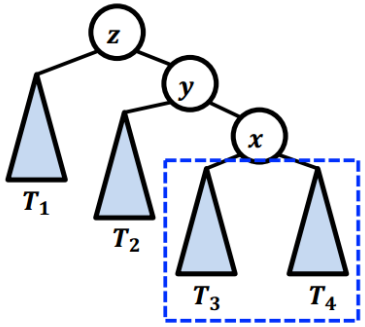
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Operations: Insertion

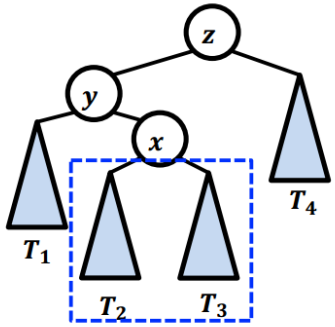
- Four cases when node z makes T unbalanced
  - [Case 1] (LL) : Insertion into the **Left** subtree of the **Left** child of node z.
  - [Case 2] (RR) : Insertion into the **Right** subtree of the **Right** child of node z.
  - [Case 3] (LR) : Insertion into the **Right** subtree of the **Left** child of node z.
  - [Case 4] (RL) : Insertion into the **Left** subtree of the **Right** child of node z.



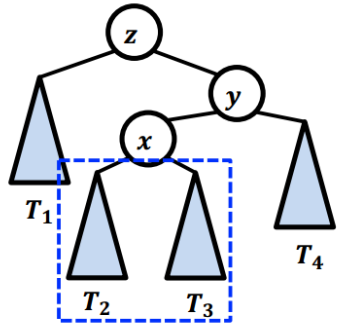
Case 1: LL



Case 2: RR



Case 3: LR

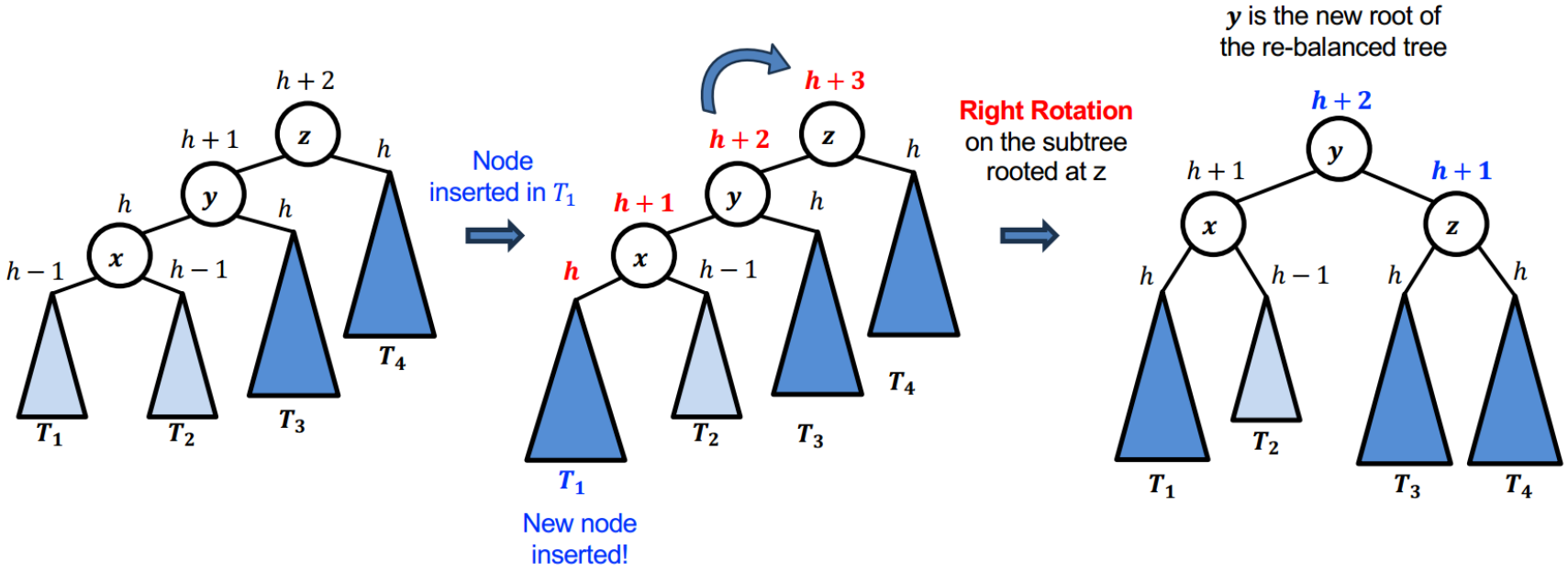


Case 4: RL

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Operations: Insertion

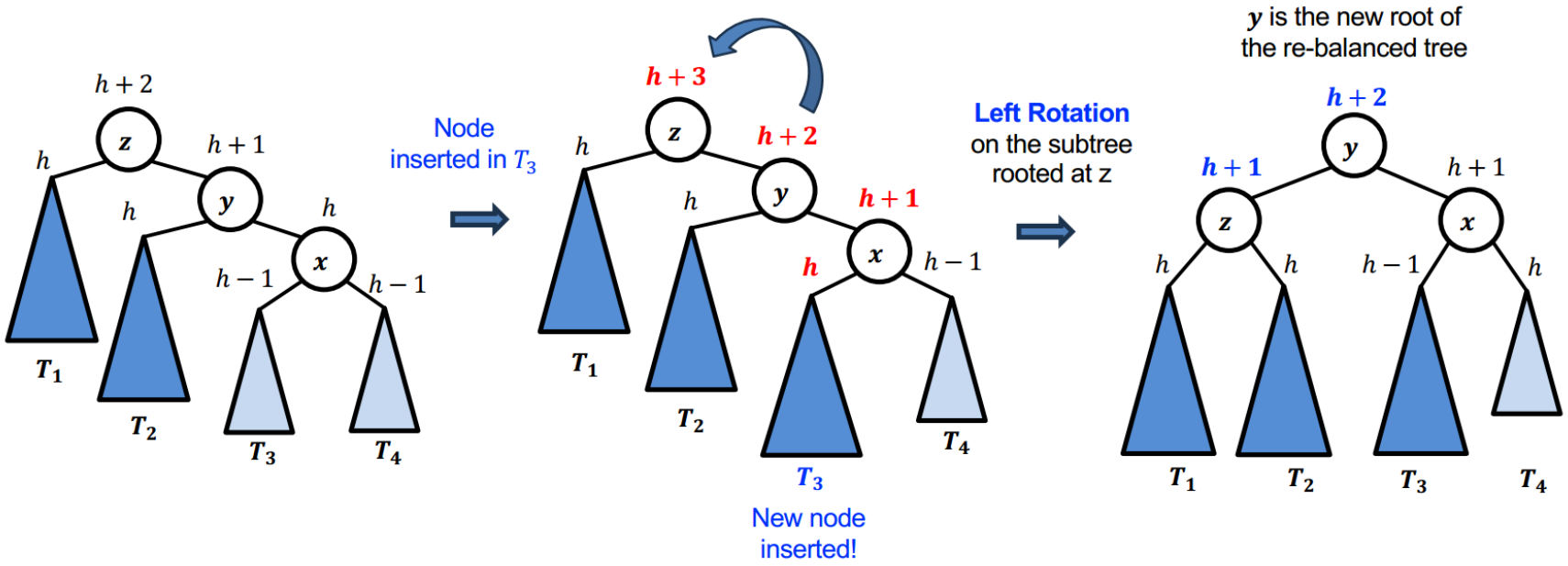
- [Case 1] (LL) : Insertion into the **Left** subtree of the **Left** child of node z.
  - Single Right Rotation



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Operations: Insertion

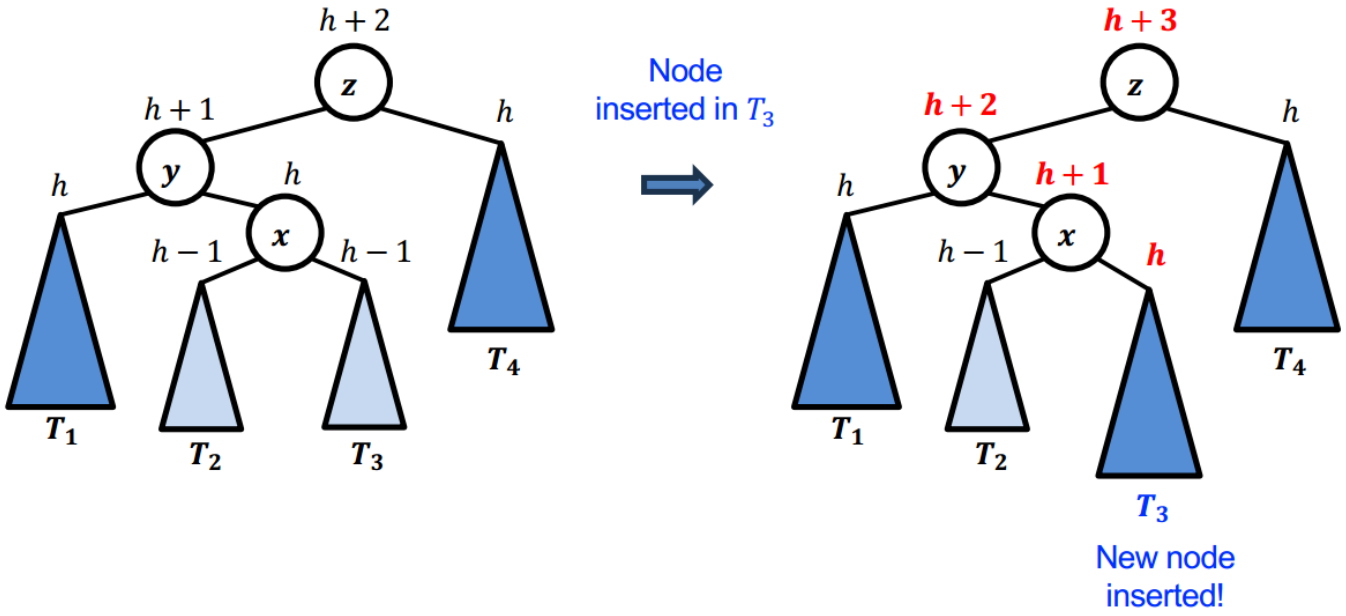
- [Case 2] (RR) : Insertion into the **Right** subtree of the **Right** child of node *z*.
  - Single Left Rotation



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Operations: Insertion

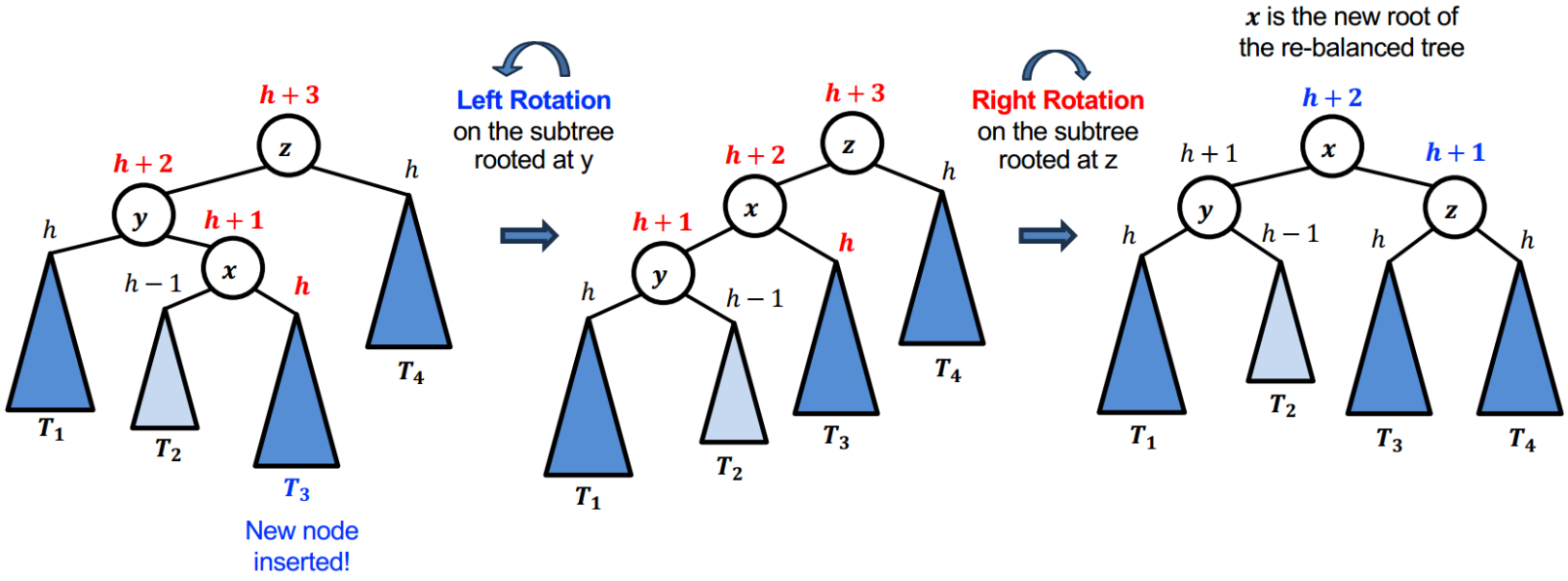
- [Case 3] (LR) : Insertion into the **Right** subtree of the **Left** child of node z.
  - Double Rotation: Left Rotation (y) + Right Rotation (z)



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Operations: Insertion

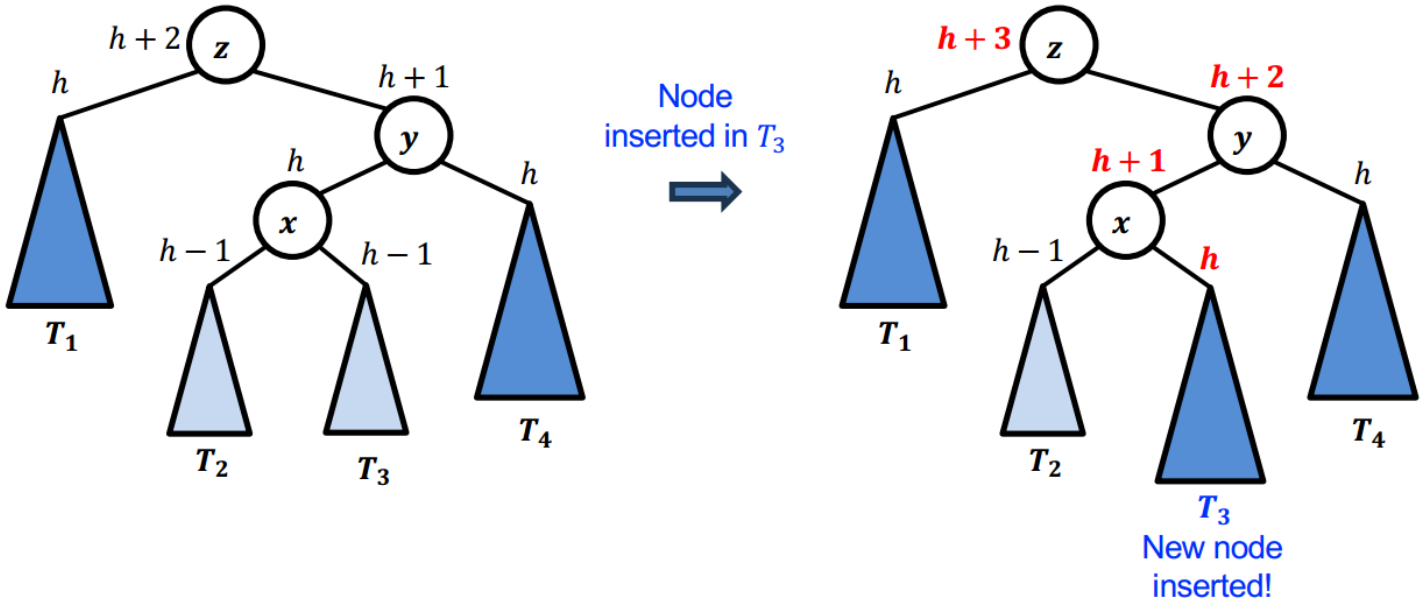
- [Case 3] (LR) : Insertion into the **Right** subtree of the **Left** child of node z.
  - Double Rotation: Left Rotation (y) + Right Rotation (z)



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Operations: Insertion

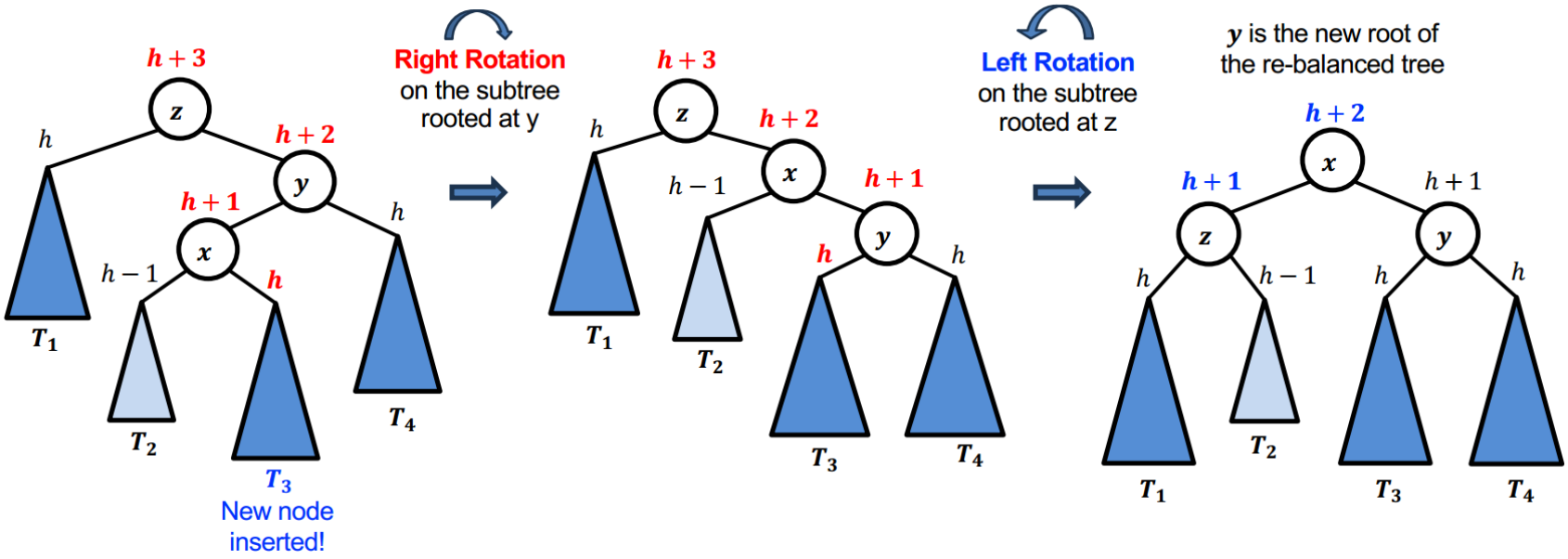
- [Case 4] (RL) : Insertion into the **Left** subtree of the **Right** child of node z.
  - Double Rotation: Right Rotation (y) + Left Rotation (z)



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Operations: Insertion

- [Case 4] (RL) : Insertion into the **Left** subtree of the **Right** child of node z.
  - Double Rotation: Right Rotation (y) + Left Rotation (z)

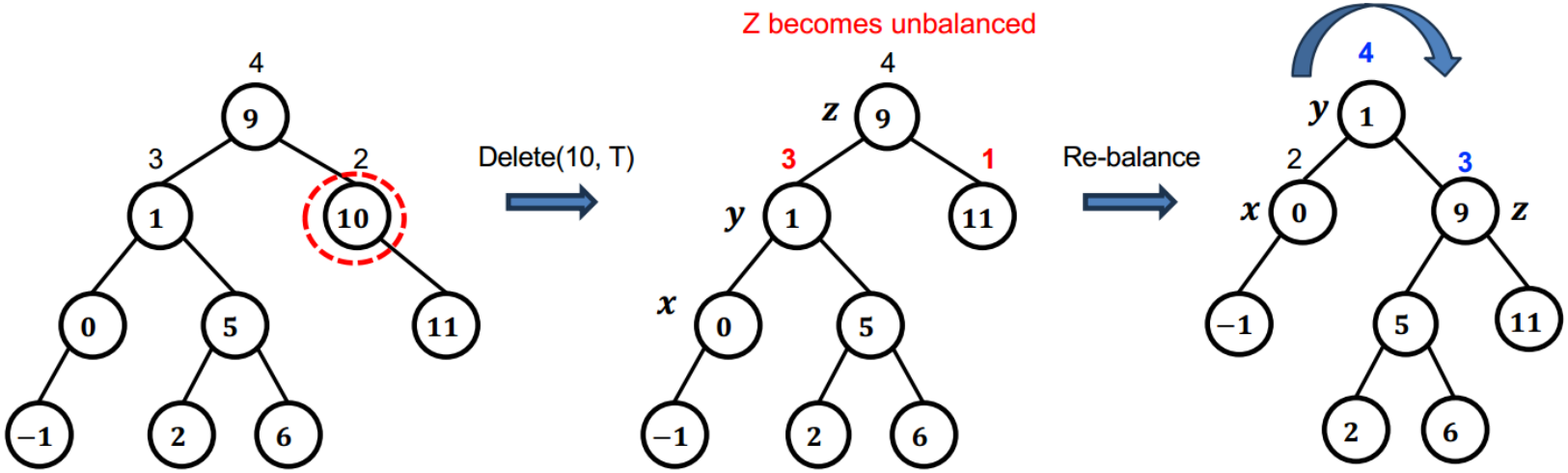


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Operations: Deletion

Deleting key w from T

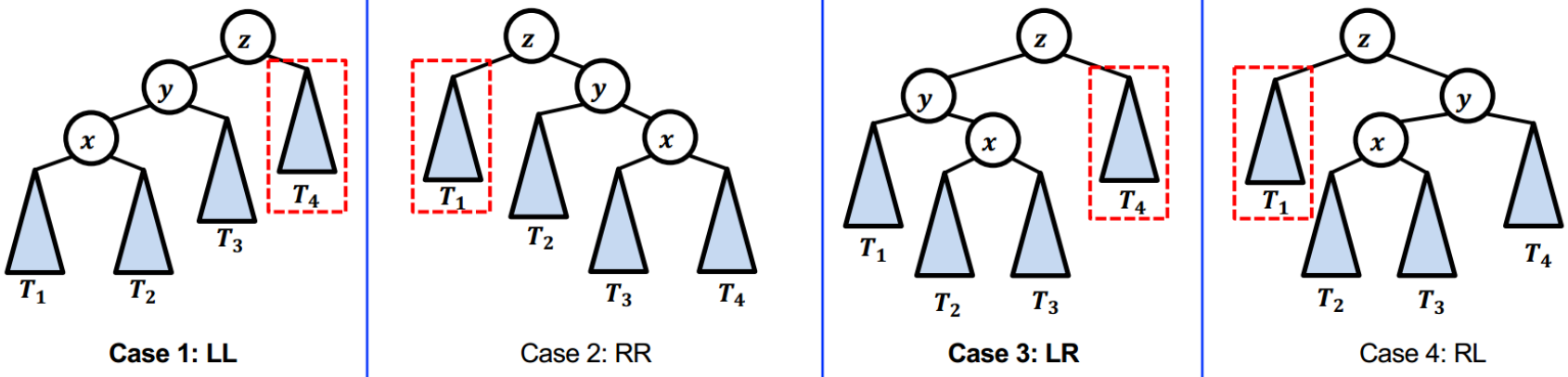
- [1] Perform standard BST deletion for the key w
  - (1) if the node has **no** child, (2) if the node has **one** child, (3) if the node has **two** children
- [2] Starting with w, find the first unbalanced node (i.e., one of the ancestors)
  - Let z be the first unbalanced node, y be the larger child of z and x be the larger child of y
- [3] Re-balance the tree by performing appropriate rotations on the subtree rooted at the z, the unbalanced node.



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Operations: Deletion

- Four cases when the node z becomes unbalanced
  - Deletion of a node in T_4 : Case 1 (LL), Case 3 (LR)
    - When the height of T_4 decreases.
  - Deletion of a node in T_1 : Case 2 (RR), Case 2 (RL)
    - When the height of T_1 decreases



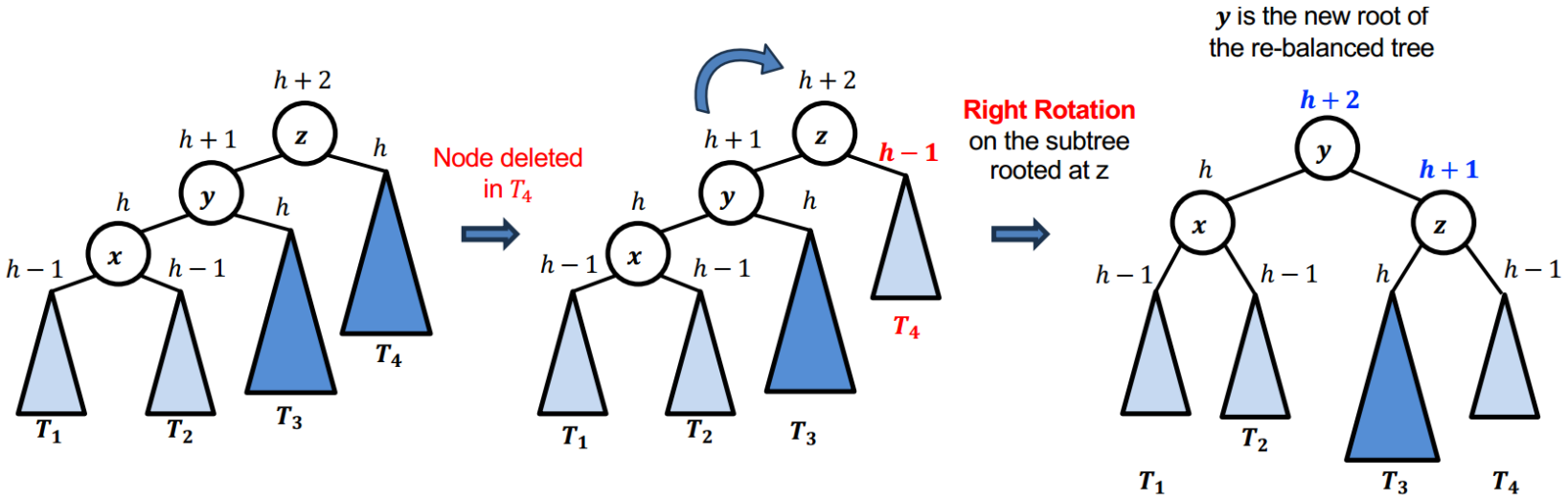
Cases 1 and 3 are symmetric to Cases 2 and 4

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Operations: Deletion

[Case 1] Deletion of a node in T_4

- = Insertion into the **Left** subtree of the **Left** child of node z
- LL case (Single rotation) : Right rotation(z)

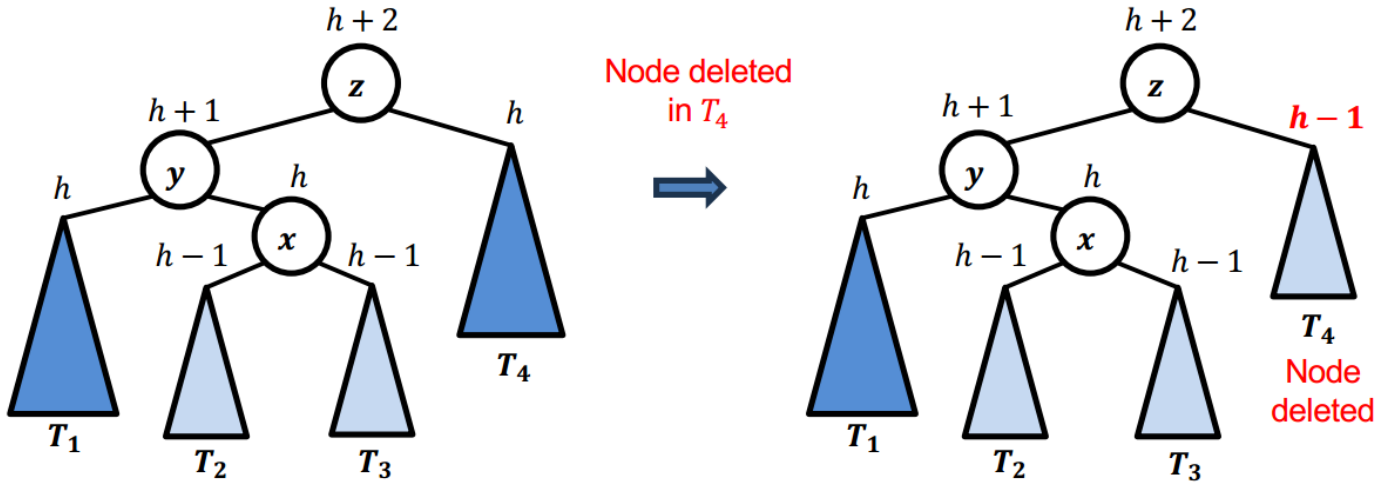


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Operations: Deletion

[Case 3] Deletion of a node in T_4

- = Insertion into the Right subtree of the **Left** child of node z
- LR case (Double rotation) : Left rotation(y) + Right rotation (z)

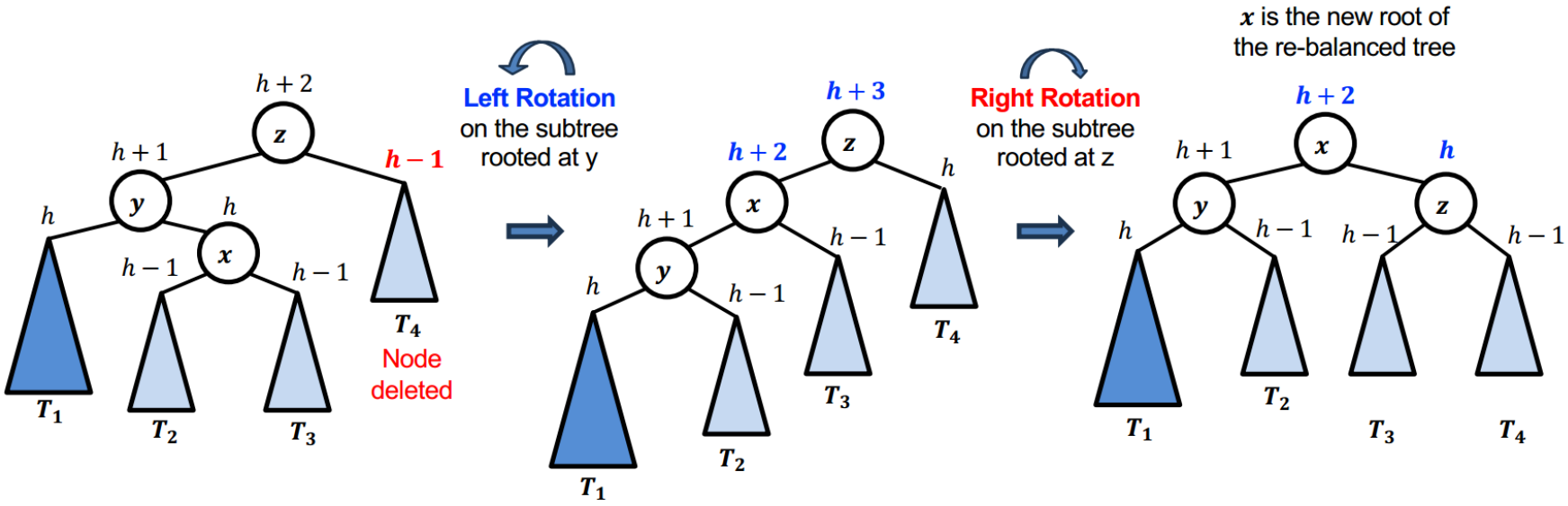


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Operations: Deletion

[Case 3] Deletion of a node in T_4

- = Insertion into the Right subtree of the Left child of node z
- LR case (Double rotation) : Left rotation(y) + Right rotation (z)



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AVL Tree

Example

- Insert(50), Insert(70), Insert(90), Insert(20), Insert(60), Insert(30), Delete(20), Insert(10), Insert(55), Insert(65), Delete(30), Insert(57), Delete(90), Delete(60), Delete(57), Delete(70), Insert(20)

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Example

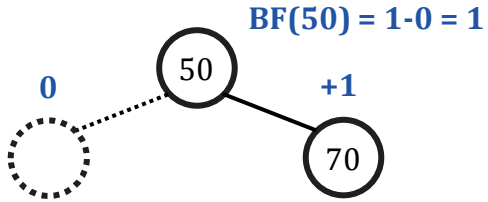
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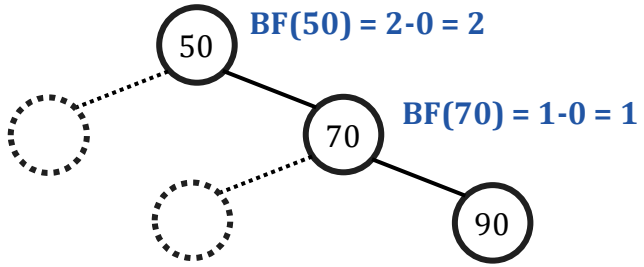
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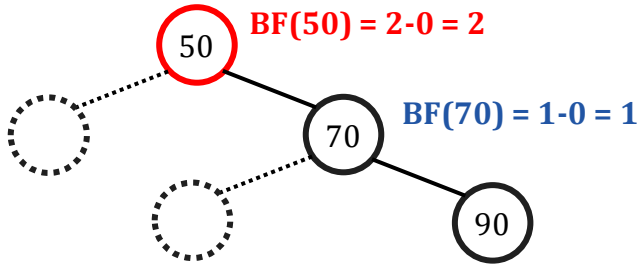
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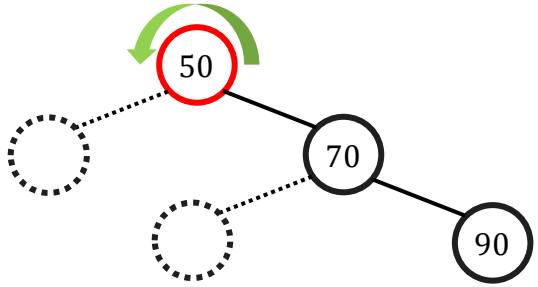
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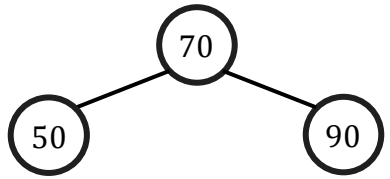
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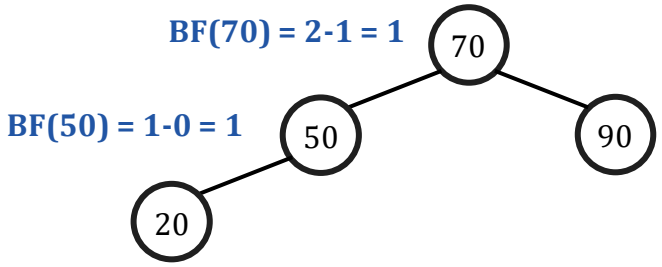


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AVL Tree

Example

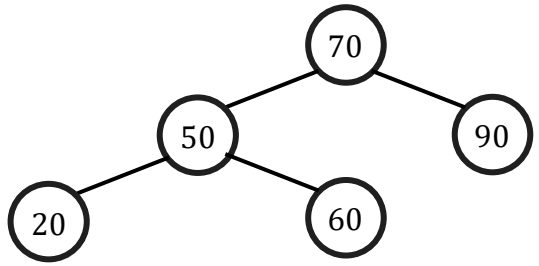
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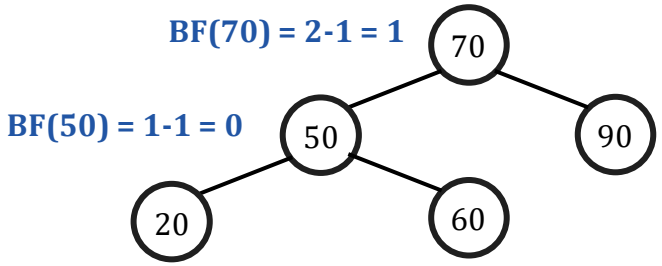
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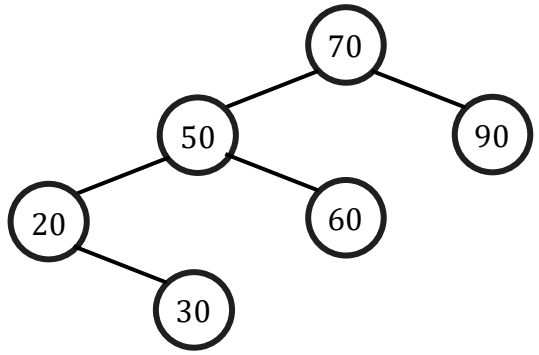


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AVL Tree

Example

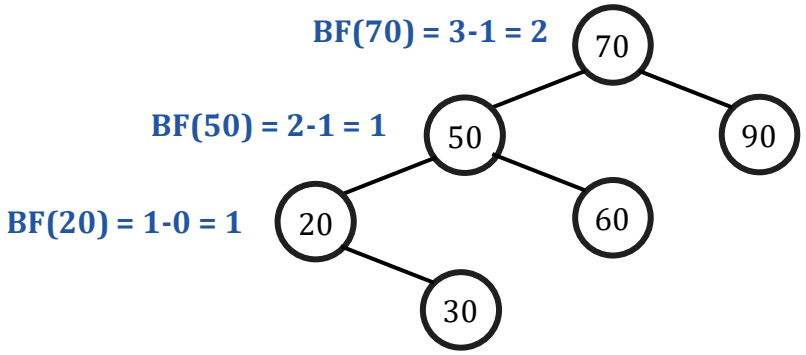
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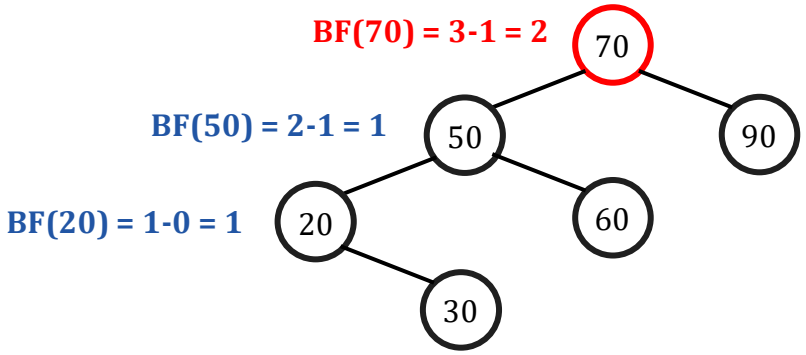
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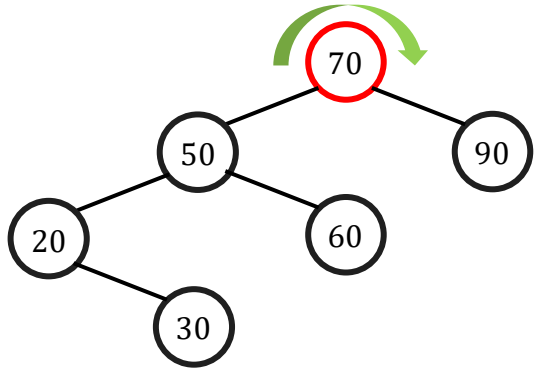
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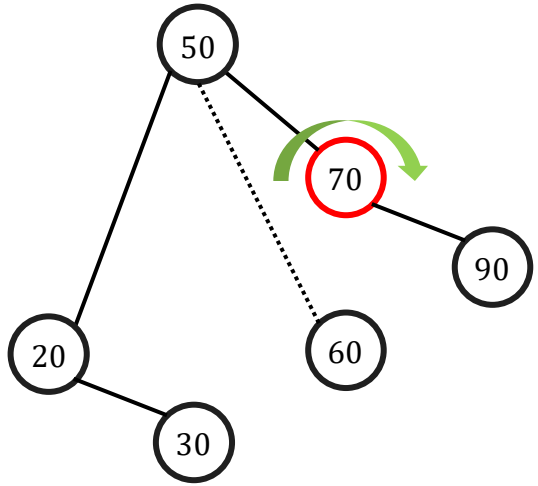
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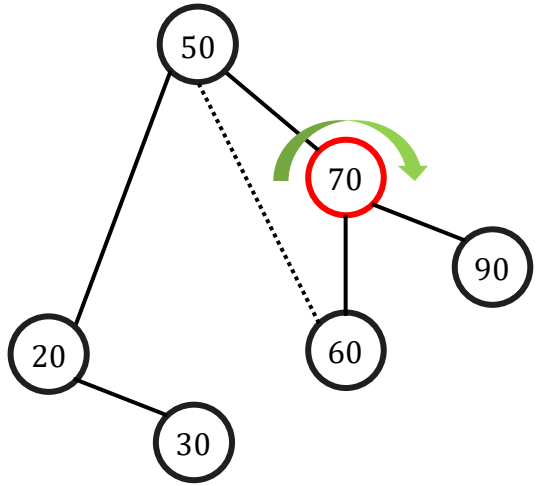
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Example

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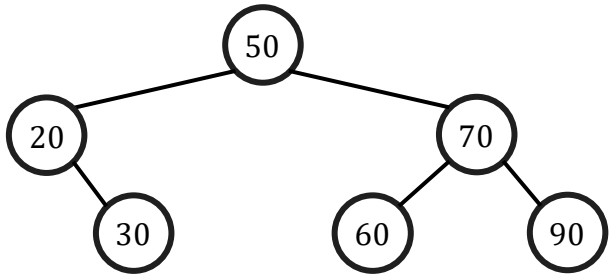


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AVL Tree

Example

- Insert(50), Insert(70), Insert(90), Insert(20), Insert(60), Insert(30), Delete(20), Insert(10), Insert(55), Insert(65), Delete(30), Insert(57), Delete(90), Delete(60), Delete(57), Delete(70), Insert(20)

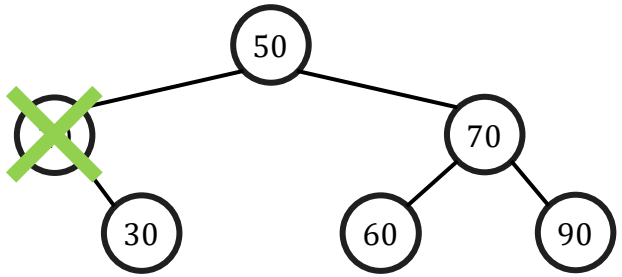


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AVL Tree

Example

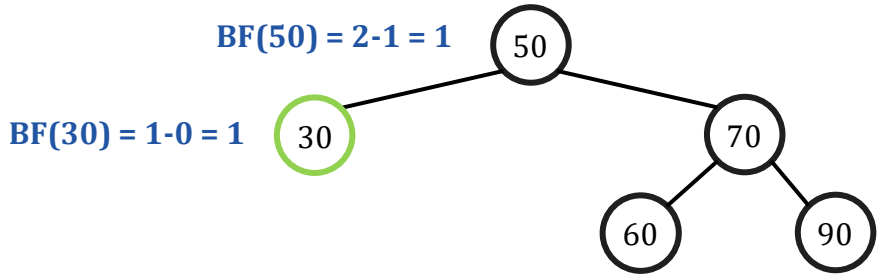
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Example

- Insert(50), Insert(70), Insert(90), Insert(20), Insert(60), Insert(30), Delete(20), Insert(10), Insert(55), Insert(65), Delete(30), Insert(57), Delete(90), Delete(60), Delete(57), Delete(70), Insert(20)

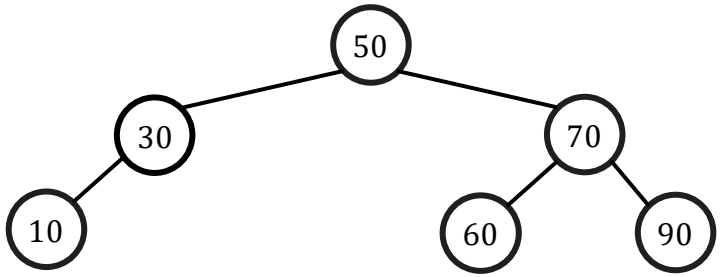


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AVL Tree

Example

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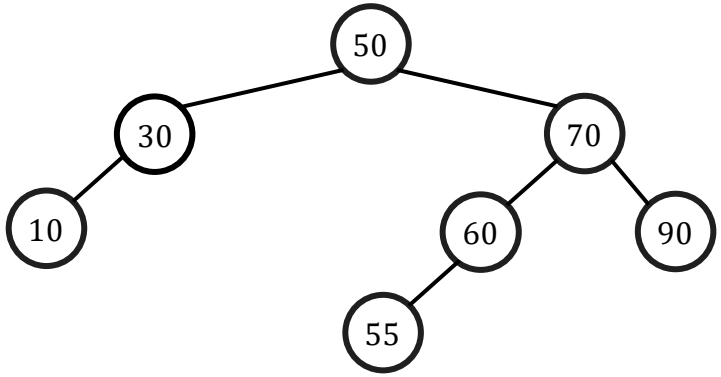


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AVL Tree

Example

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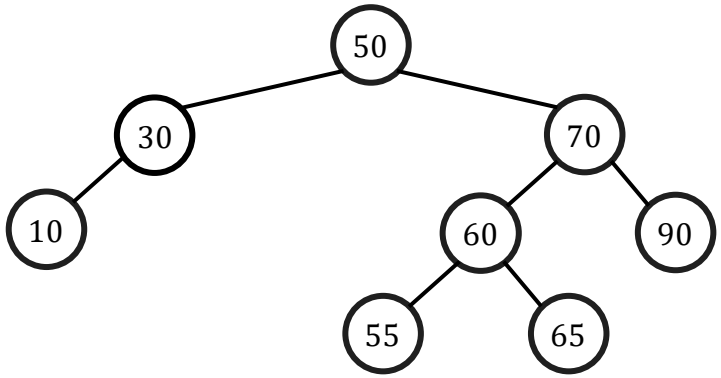


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AVL Tree

Example

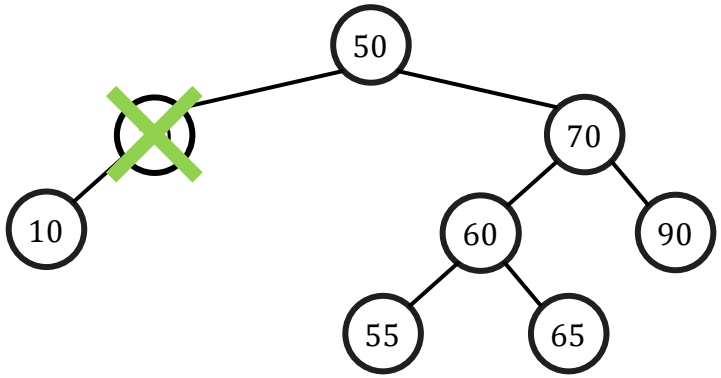
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Example

- Insert(50), Insert(70), Insert(90), Insert(20), Insert(60), Insert(30), Delete(20), Insert(10), Insert(55), Insert(65), Delete(30), Insert(57), Delete(90), Delete(60), Delete(57), Delete(70), Insert(20)

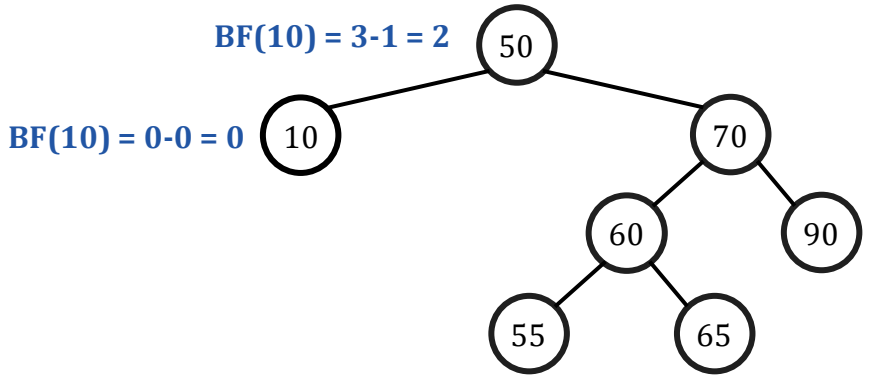


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AVL Tree

Example

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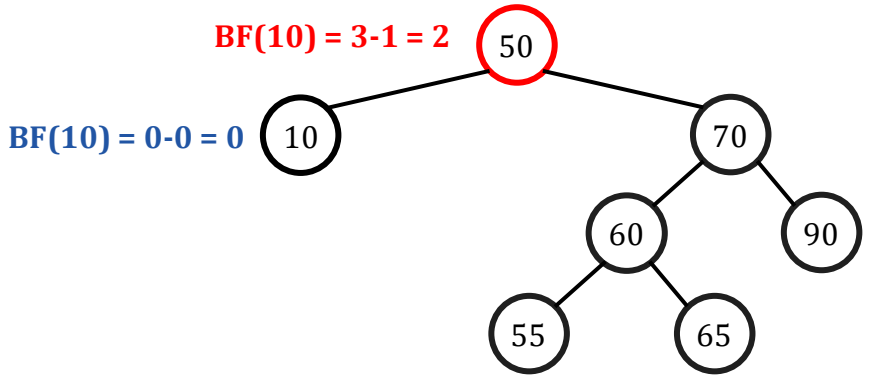


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AVL Tree

Example

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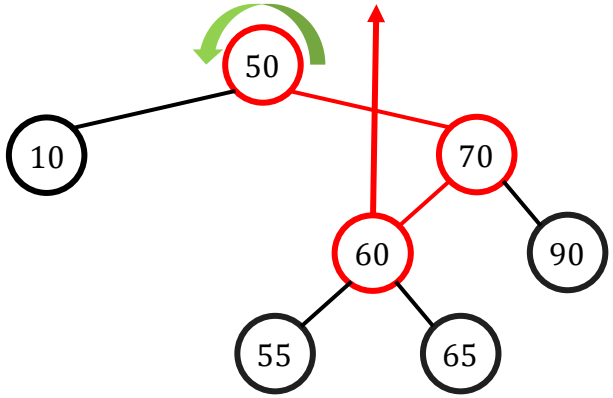


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AVL Tree

Example

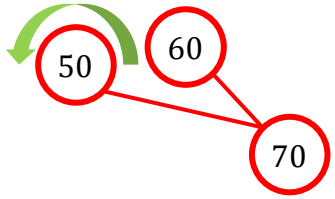
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Example

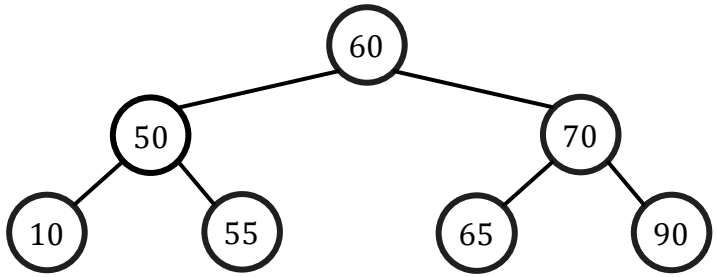
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Example

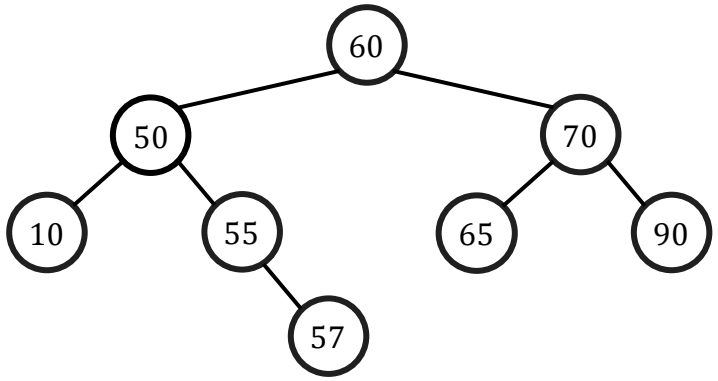
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Example

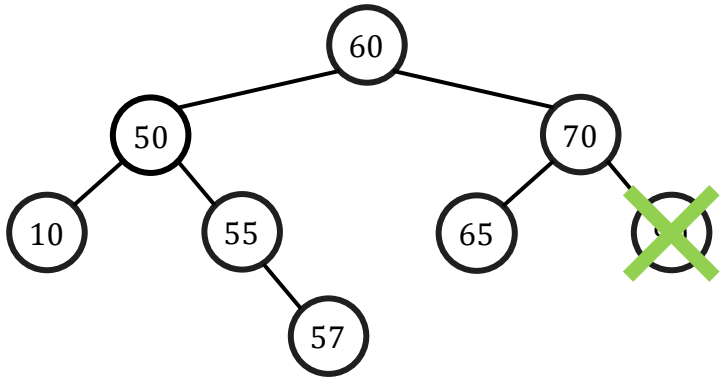
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Example

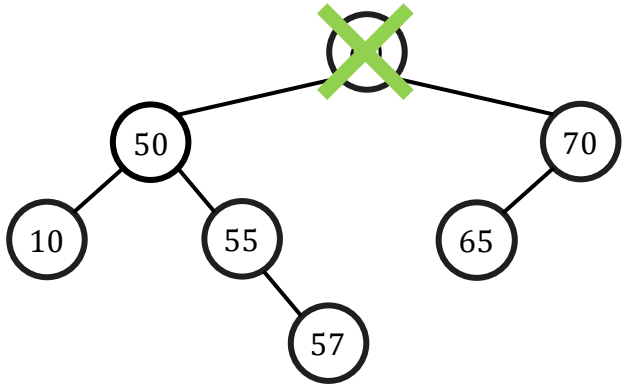
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Example

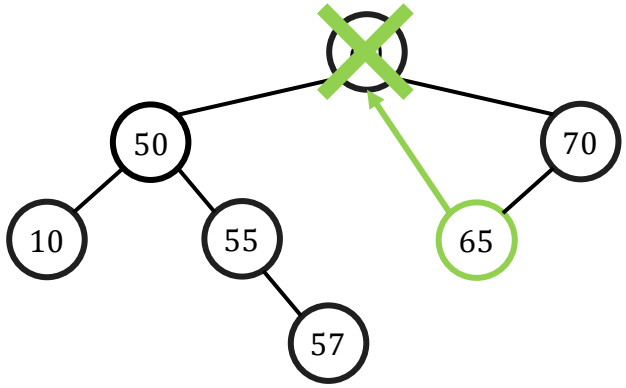
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Example

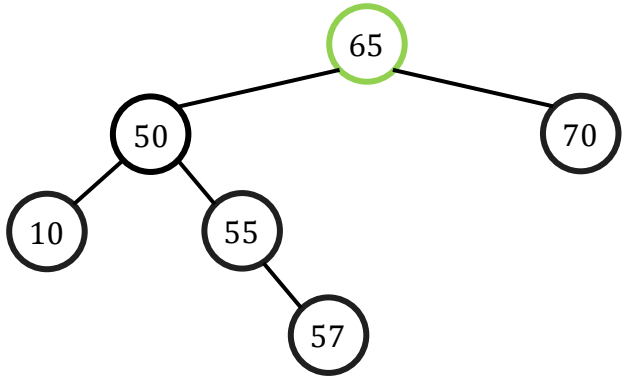
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Example

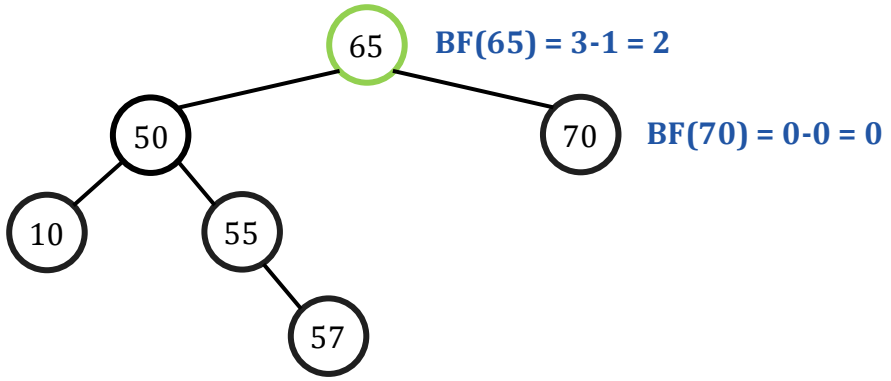
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Example

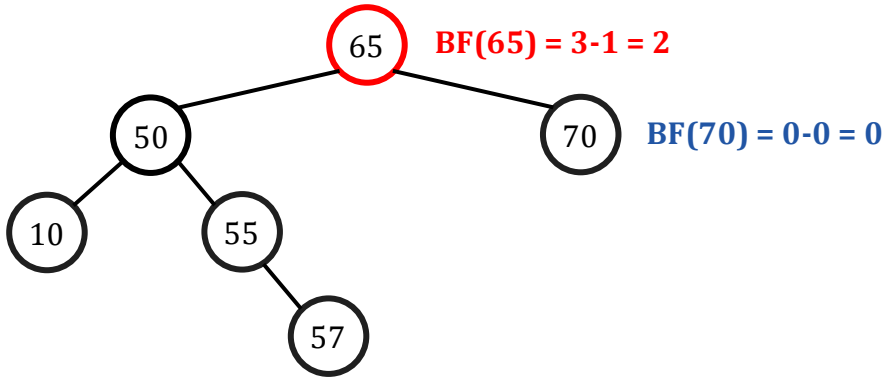
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Example

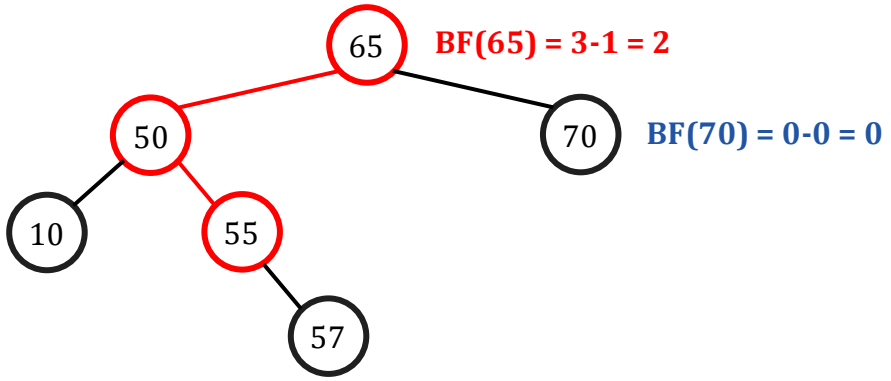
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Example

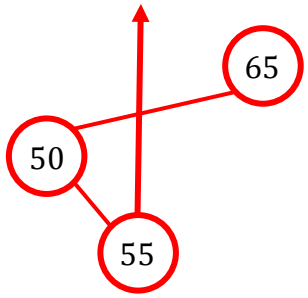
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Example

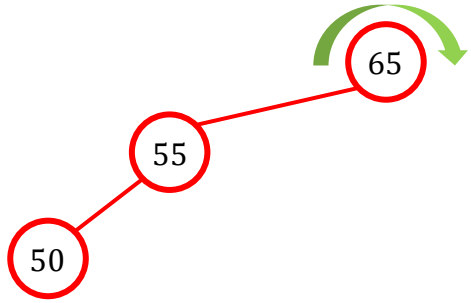
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Example

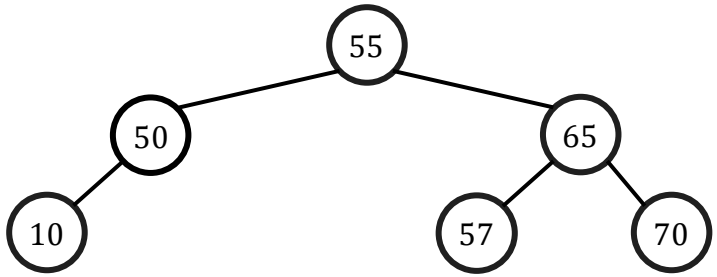
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Example

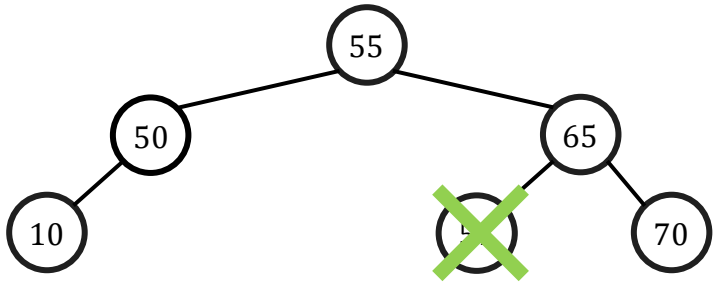
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Example

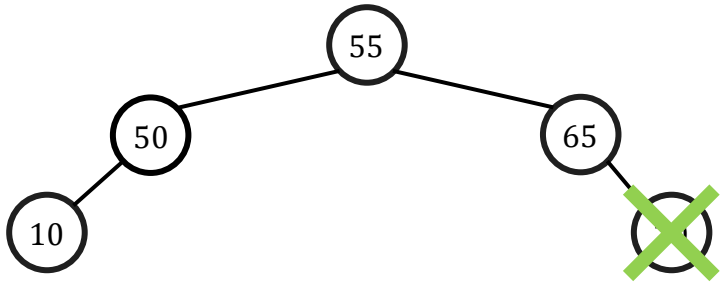
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Example

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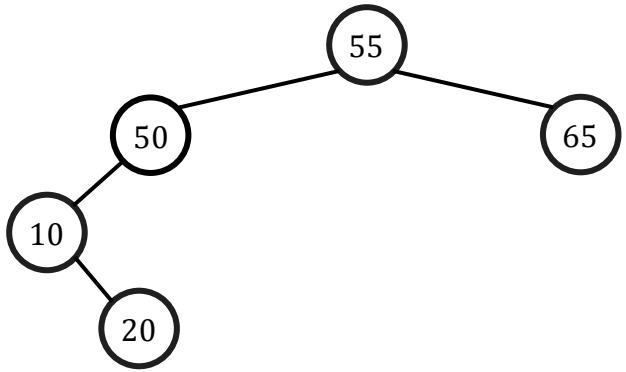


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AVL Tree

Example

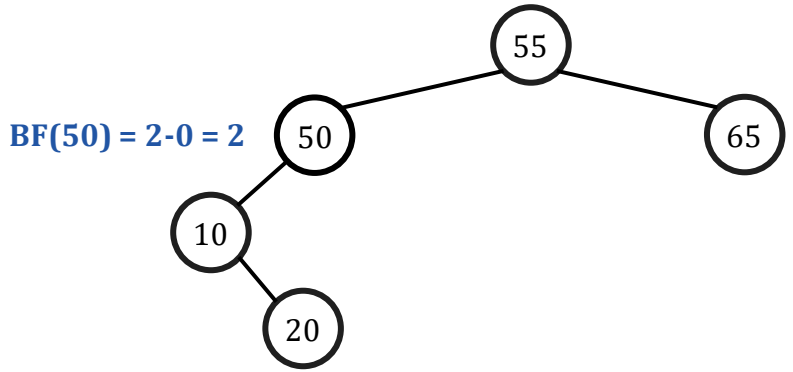
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Example

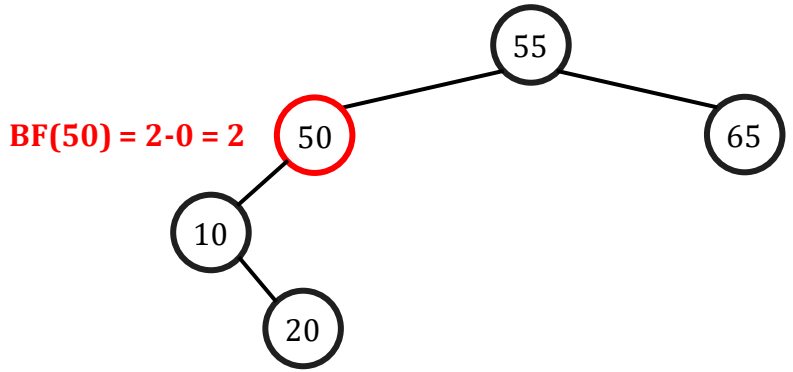
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Example

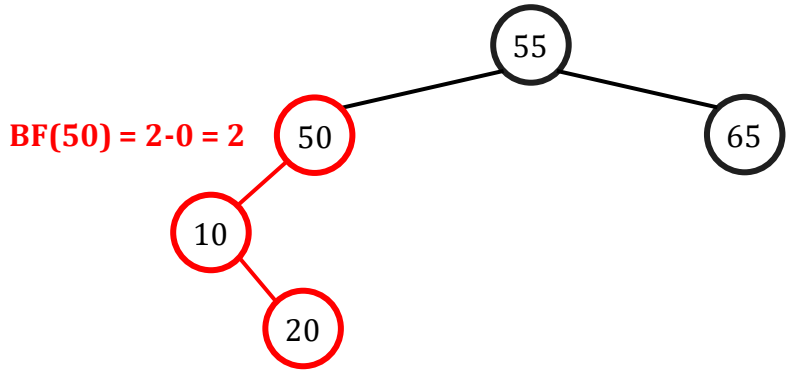
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Example

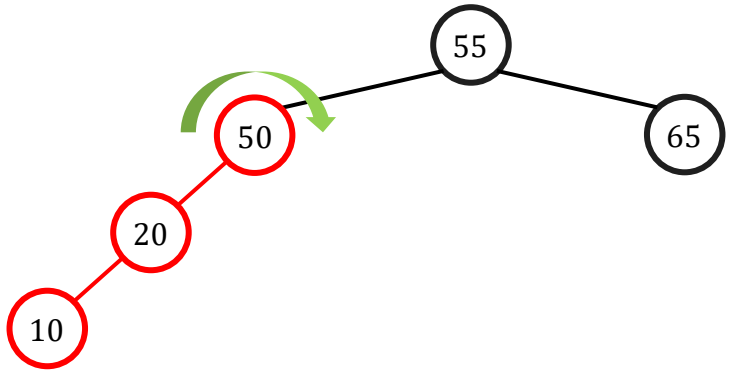
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Example

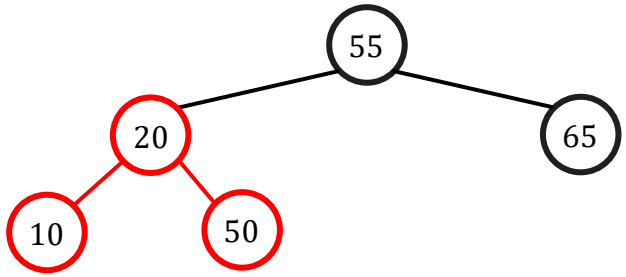
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Example

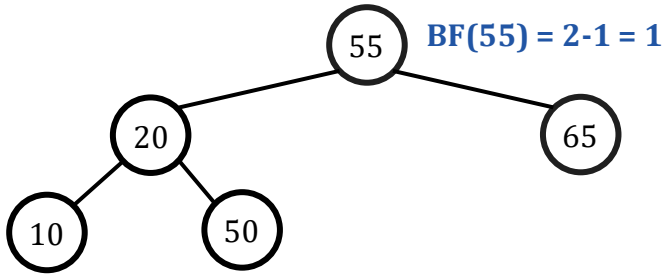
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Example

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Thank you

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Office : 310-430